Fuses Types

Lecture 2

Low voltage fuses

- **1- Rewirable Fuse**
- The most famous kit-kat fuse (also known as rewirable fuse) is mostly used in industries and home electrical wiring for small current applications in Low Voltage (LV) systems.
- A rewirable fuse contains 2 basic parts. The inner fuse element as a fuse carrier made of tinned copper, Aluminum, Lead etc. and the base is made of porcelain having the IN and OUT terminals, which is used in series with the circuit to protect.
- It is made from a porcelain or ceramic insulator.

Advantage

The wire is enclosed in a cartridge-type container.

 The wrong size of fuse cannot be fitted since it is with a different size for a different current.

 The fuse wire does not deteriorate and is more reliable.

 It can be rewired easily in case it is blown due to short circuit or over current which melts the fuse elements.

2- Cartridge Fuse.

- ***D – Type Cartridge Fuse:***

D-Types fuse contains an adapter ring, base, cap and cartridge. The fuse base is connected to the fuse cap, where the cartridge is located inside. The circuit is completed when the tip of the cartridge makes contact with the fuse link conductor.

- **HRC (High Rupturing Capacity) Fuse or Link Type Cartridge Fuse**

High Rupturing Capacity (HRC) fuses, also known as cartridge or link type fuses, including their construction, operation, and applications.

A link-type cartridge fuse is a cylindrical fuse with a metal fuse link inside that melts to break a circuit during an overcurrent. "Link type" refers to the fuse's design, which includes a sacrificial fuse link that melts, while "cartridge" refers to the cylindrical shape that houses the link.

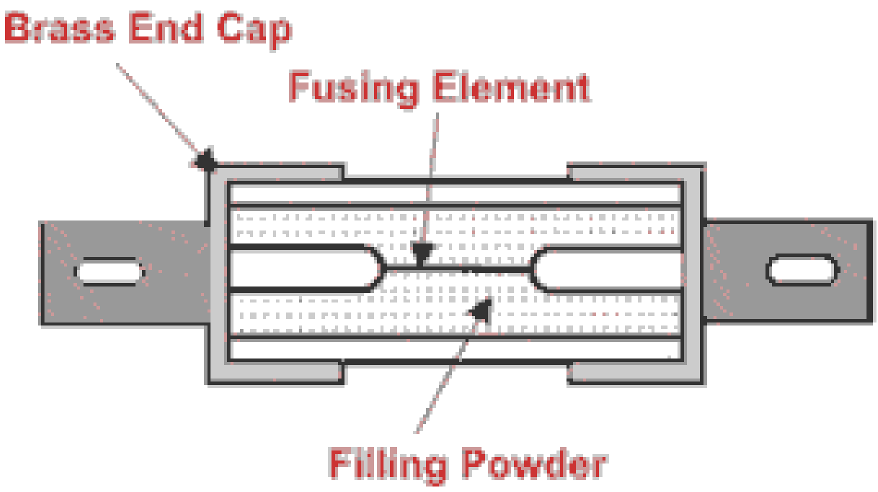
Advantage

- The wire is enclosed in a cartridge-type container.
- The wrong size of fuse cannot be fitted since it is with a different size for a different current.
- The fuse wire does not weaken and is more reliable.

3- High Rupturing Capacity (HRC)

It is a cartridge-type with a silver element connected between two end-contacts of a ceramic tube filled with a special quartz powder.

This type covered various types of HRC fuses, such as DIN type, NH Type, Blade Type, Liquid Type HRC Fuse, and Expulsion Type HV Fuse, along with their advantages and disadvantages.



Construction of HRC Fuse

Advantage:

- This type of fuse is very reliable in performance and does not weaken, and has a high speed.
- When the fuse silver blows (which rapidly melts), produced with the filling powder, high-resistance materials are formed in of operation and the path of the arc, causing it to be extinguished.
- Ability to safely interrupt short circuit currents of much higher values (higher rupturing capacity).
- Low cost
- Wire may be easily available.

Disadvantage

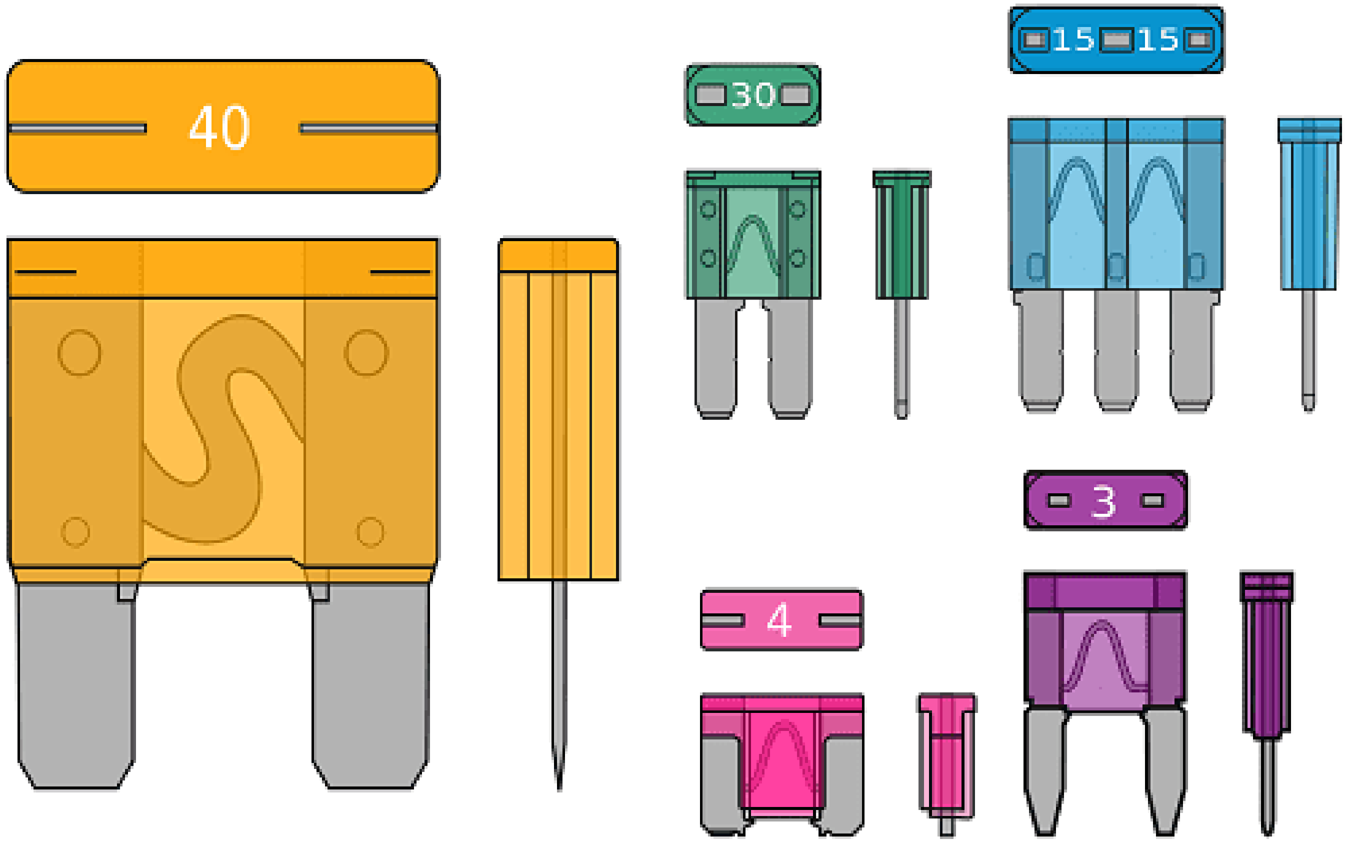
 The wrong size of wire to be fitted in the fuse cause wrong operation at high current circuits, which may be dangerous for the circuit protected and not adequate for electrical arc extinguish

4- Automotive, *Blade Type & Bolted Type Fuses*

These types of fuses (also known as **spade** or **plug-in fuses**) come in a plastic body and two metal caps to fit in the socket.

Mostly, they are used in **automobiles** for wiring and short circuit protection. Fuse Limiters, Glass Tube (also known as Bosch Fuse) are widely used in automotive industries. The rating of automobile fuses are low as 12V to 42V.

In bolted types of fuses, the base of the fuse contacted directly to the base of the fuse, same like HRC Fuses.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30
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5- SMD Fuses (Surface Mount Fuse), Chip , Radial, and Lead Fuses

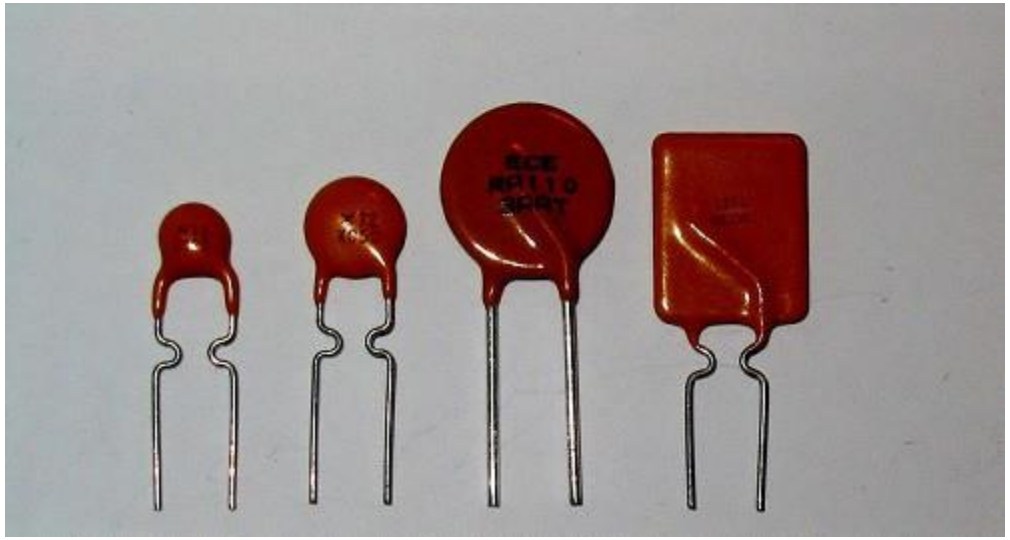
SMD Fuses (Surface Mount Device and the name derived from SMT = Surface Mount Technology) are chip types of fuses (also known as electronic fuse) are used in DC power applications like Hard drives, DVD players, Camera, cell phones etc. Space plays an important role because SMD fuses are very tiny in size and hard to replace as well.

Below are some additional types of SMD Fuses and Leaded fuses.

- Slow – Blow Chip Fuses
- Fast Acting Chip Fuses
- Very Fast Acting Chip Fuses
- Pulse Tolerant Chip Fuses
- High Current Rated Chip Fuses
- Telecom Fuses
- Through-hole styles fuses
- Radial Fuse
- Lead Fuse
- Axial Fuse

6- Resettable Fuses

Resettable fuse is a device that can be used multiple times without replacing it. They open the circuit when an over current event occurs, and after some specific time, they reconnect the circuit. Application of resettable fuses is overcome where manually replacing fuses is difficult or almost impossible, e.g. fuse in the nuclear system or in an aerospace system.



7- Dropout Fuse:

A drop-out fuse is an expulsion-type fuse to protect the transformers. When the fuse element melts, it falls due to gravity, hence providing additional isolation.

Drop out fuse

8- Switch Fuse:

A switch fuse is used for low and medium-voltage circuits. They can safely break depending upon the rating currents of the order of 3 times the load current.

A switch fuse, or fused switch, is a single unit that combines a manually operated switch for isolating an electrical circuit with built-in overcurrent protection through a fuse. It is used to safely turn off power and protect circuits from overloads or short circuits in commercial, industrial, and residential settings.


LSeries
LSeries
LSeries
32A
32A
32A

8- Striker Fuse:

This type of fuse has a mechanical indicator or striker pin that protrudes through the fuse cap upon operation of the fuse. This provides visual identification of a blown fuse and acts as a trigger for external devices.

It can be used for short circuit protection of medium voltage motors.

Striker fuse (Source: mc-mc.com)

High Voltage Fuses

- High Voltage (HV) fuses are used in power systems to protect the power transformer, distribution transformers, and instrument transformers etc, where circuit breakers may be inadequate.
- HV fuses are designed for high-voltage applications, typically exceeding 1500V and ranging up to 13kV.
- The element of a high-voltage fuse is generally made of copper, silver, or tin.
- The fuse link chamber may be filled with boric acid in case of expulsion-type HV (High Voltage) Fuses.
- Same characteristics and operation as LV fuse but differing in size and shape.

High Voltage Fuses types

- **1- Cartridge Fuse**.
- Cartridge fuses are used to protect electrical appliances such as motors, air-conditioners, refrigerators, pumps, etc, where high voltage ratings and currents are required. They are available up to 600A and 600V AC and widely used in industries, commercial, as well as home distribution panels.
- There are two types of Cartridge fuses.
- 1. **General-purpose fuse** with no time delay.
- 2. **Heavy-duty cartridge fuses** with a time delay.

Both are available in 250V AC to 600V AC, and their rating can be found on the end cap or knife blade.

Cartridge type HV HRC fuse

2- Expulsion fuses

An expulsion type HRC (High Rupturing Capacity) fuse is a high-voltage fuse that protects circuits by venting exhaust gases during a fault. These fuses, also called boric-acid fuses, use a fuse link inside an arc-extinguishing tube to interrupt high fault currents. When the fuse blows, the resulting arc is extinguished as the gases are expelled from the tube.

Large power fuses use fusible elements made of silver, copper or tin to provide stable and predictable performance. High voltage *expulsion fuses* surround the fusible link with gas-evolving substances, such as boric acid. When the fuse blows, heat from the arc causes the boric acid to evolve large volumes of gases. The associated high pressure (often greater than 100 atmospheres) and cooling gases rapidly reduce the resulting arc. The hot gases are then explosively expelled out of the end(s) of the fuse. Such fuses can only be used outdoors.

These types of fuses may have an impact pin to operate a switch mechanism, so that all three phases are interrupted if any one fuse blows.

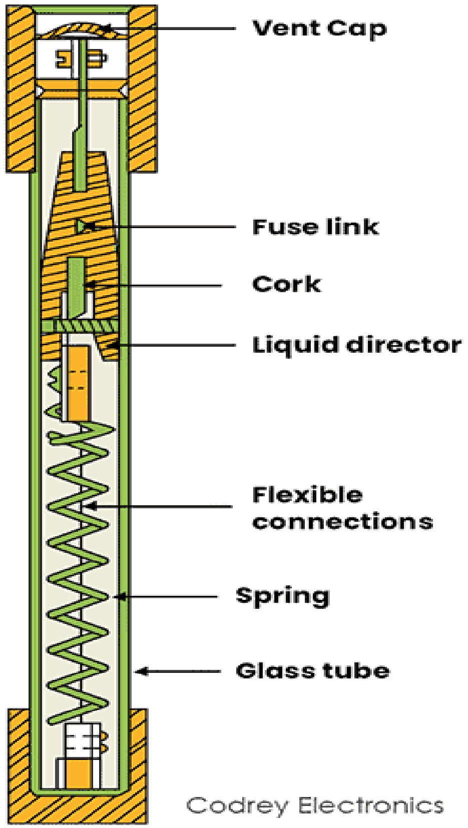
High-power fuse means that these fuses can interrupt several kilo amperes. Some manufacturers have tested their fuses for up to 63 kA short-circuit current.

Expulsion type HV HRC fuse

3- Liquid type HRC

A liquid-type High Rupturing Capacity (HRC) fuse is a high-voltage fuse filled with an arc-reducing liquid, typically carbon chloride or transformer oil, to defeat the electrical arc when the fuse element melts during a fault. The liquid rapidly cools the arc, preventing further current flow and protecting equipment like transformers and circuit breakers in high-voltage systems.

A liquid fuse (for high currents) consists of a glass tube filled with carbon chloride and sealed with brass caps at both ends. The fuse wire covers sealing at one end and fixing by strong phosphor bronze spiral spring at another end of the glass tube. The liquid acts as an arc extinguishing medium.


Vent Cap
Fuse link
Cork
Liquid director
Flexible connections
Spring
Glass tube
Codrey Electronics